PHYS4055

Atomic Physics and Quantum Optics

Open-book examination: four-part course overview

Source-reviewed revision edition 13 November 2026

Purpose. A compact conceptual reference: each topic explains the physical system, its governing equations, the assumptions and how to use the equations in calculations. Use this with the separate worked-example and past-exam books. It is not a substitute for the original slides when a derivation depends on a figure or convention.

The four parts of the course

Course part	Main questions	Pages
I. Quantum optics	What is a quantum field mode? How do Fock/coherent, thermal and squeezed states differ? How do beam splitters, photodetectors, quadratures and quasiprobabilities describe optical experiments?	2–4
II. Applications of optics	How do photonic qubits and entanglement enable quantum communication? What do open-system noise, conditional measurements, atom–cavity coupling and quantum metrology accomplish?	5–9
III. Atomic physics	How do we calculate atomic orbitals and energies? How do angular-momentum addition, exchange symmetry, fine/hyperfine interactions and external fields determine the spectrum?	10–12
IV. Applications of atomic physics	Why do spectral lines have selection rules and lifetimes? How can lasers cool atoms? How do atoms test fundamental symmetries and operate as extremely accurate clocks?	13–15

How to use this booklet during the exam

1. **Choose a physical model.** Number-state optics, Gaussian optics, a quantum channel, a two-level atom, or a coupled angular-momentum multiplet have different natural bases.
2. **Write conventions.** For optics we use $x = a + a^\dagger$, $p = i(a^\dagger - a)$, so vacuum variance is 1. For atomic physics state l, s, L, S, J, I, F and the basis.
3. **Start from the boxed equation and the method box.** A named model alone is not an answer: show enough algebra to connect the question's given quantities to the requested observable.
4. **Perform a physical check:** conservation of probability, state normalisation, commutators, degeneracy sums, operator parity, triangle rules, limiting cases, dimensions and units.

Source audit and notation

The underlying lecture slides were reviewed across Weeks 1–6 (optics and applications), Weeks 8–9 (atomic foundations), Week 10 (spectroscopy/cooling), Week 12 (symmetry tests) and Week 13 (clocks), alongside the available lecture transcripts. Individual pages cite slide packs and page ranges. Slide/text extraction can misread equations, particularly in annotated material: the original PDFs remain authoritative. The **source map** next to this booklet records coverage and known issues.

This overview is educational revision material, not an official formula sheet or a prediction of exam questions. Date of preparation: 13 November 2026.

Part I. Quantum optics 2 / 15

1. The electromagnetic field becomes a set of oscillators

A classical Maxwell field can be decomposed into independent spatial/polarisation modes. Each mode has a harmonic-oscillator Hamiltonian. Quantising a mode promotes its amplitude and conjugate momentum to operators $[\hat{q}, \hat{p}] = i\hbar$, giving

$$\boxed{\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + \tfrac{1}{2})}, \qquad [\hat{a}, \hat{a}^\dagger] = 1, \quad \hat{n} = \hat{a}^\dagger\hat{a}.$$

The number (Fock) states obey

$$\hat{a}\,|n\rangle = \sqrt{n}\;|n-1\rangle\,, \qquad \hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\;|n+1\rangle\,, \qquad \hat{n}\,|n\rangle = n\,|n\rangle\,.$$

A photon is one excitation of a *mode*, not a tiny object appended to a classical wave. For several independent modes, $\hat{H} = \sum_j \hbar\omega_j(\hat{n}_j + \frac{1}{2})$ and $|n_1, n_2, \ldots\rangle = |n_1\rangle \otimes |n_2\rangle \otimes \cdots$. Mode labels (frequency, wavevector, polarisation, path) therefore matter.

2. Coherent states: a quantum version of a classical field

A coherent state is an eigenstate of annihilation:

$$\boxed{\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle}, \qquad |\alpha\rangle = \hat{D}(\alpha)\,|0\rangle = e^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\,|n\rangle\,, \quad \hat{D}(\alpha) = e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}.$$

Thus $\langle\hat{n}\rangle = |\alpha|^2$, while the optical phase is encoded in $\arg\alpha$. The identity $\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha$ lets you evaluate expectation values using the vacuum.

> **Typical calculation:** for a coherent input, expand in Fock states if the question asks for photon probabilities; use $\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle$ for normally ordered moments; use $\hat{D}^\dagger\hat{a}\hat{D}$ for field quadratures.

3. Why propagation and modes matter

An infinitely monochromatic plane wave has no finite temporal envelope. A physical pulse is a superposition of frequency modes. The free evolution $\hat{a}_j(t) = \hat{a}_j(0)e^{-i\omega_j t}$ rotates each mode's phase. Which modes are combined by optics determines whether they interfere.

Lecture sources: W1L1 pp. 16–41; W1L2 pp. 3–29; W1L3 pp. 2–3; quantum-optics lecture notes, sections 1–2.

Part I. Quantum optics 3 / 15

1. Mixed states and photon-number statistics

A density operator $\rho = \sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right|$ represents either a pure state ($\rho^2 = \rho$) or a statistical mixture. Photon counting gives $p_n = \langle n|\,\rho\,|n\rangle$ and

$$\bar{n} = \mathrm{Tr}(\rho\hat{n}), \quad (\Delta n)^2 = \langle \hat{n}^2\rangle - \bar{n}^2, \quad Q_{\mathrm{M}} = \frac{(\Delta n)^2 - \bar{n}}{\bar{n}}.$$

State	Number distribution / moments	Statistics
$\left\|n\right\rangle$	$p_m = \delta_{mn}$, variance 0	sub-Poissonian for $n > 0$
$\left\|\alpha\right\rangle$	$p_n = e^{-\bar{n}}\bar{n}^n/n!$, variance $\bar{n}$	Poissonian
thermal	$p_n = \bar{n}^n/(1+\bar{n})^{n+1}$, variance $\bar{n} + \bar{n}^2$	super-Poissonian

For thermal light, $\bar{n} = [\exp(\hbar\omega/k_B T) - 1]^{-1}$. Its density matrix is diagonal in the number basis; a coherent superposition and a mixture with the same p_n need not be equivalent.

2. A beam splitter is a two-mode unitary

With intensity transmissivity η, choose the real convention

$$\boxed{\begin{pmatrix}\hat{a}'\\ \hat{b}'\end{pmatrix} = \begin{pmatrix}\sqrt{\eta} & \sqrt{1-\eta}\\ -\sqrt{1-\eta} & \sqrt{\eta}\end{pmatrix}\begin{pmatrix}\hat{a}\\ \hat{b}\end{pmatrix}.}$$

Other conventions attach phases; probabilities agree if used consistently. The transformation conserves total photon number and commutators. For $|1,0\rangle$ input, the output has probabilities $P(1,0) = \eta,\ P(0,1) = 1-\eta$. For a coherent input plus vacuum, the output remains a product of coherent states with transformed amplitudes.

> **Multi-photon method:** write $|n,m\rangle = (\hat{a}^\dagger)^n(\hat{b}^\dagger)^m\,|0,0\rangle\,/\sqrt{n!m!}$. Transform *creation operators*, expand, then collect normalised Fock states. For two indistinguishable photons at a balanced splitter the $|1,1\rangle$ coincidence amplitude cancels (HOM bunching).

3. Detection

An ideal number-resolving detector measures $\{\left|n\right\rangle \left\langle n\right|\}$; a threshold detector distinguishes vacuum from one-or-more photons. Real efficiency and dark counts alter the outcome probabilities. Balanced homodyne uses a strong coherent local oscillator and the difference photocurrent to measure a chosen field quadrature rather than photon number.

Lecture sources: W2L1 pp. 3–20; W2L2 pp. 3–17; W2L3 pp. 5–17; W3L3 pp. 5–9.

Part I. Quantum optics 4 / 15

1. Quadratures and optical phase space

The course uses the dimensionless quadrature convention

$$\boxed{\hat{x} = \hat{a} + \hat{a}^\dagger, \quad \hat{p} = i(\hat{a}^\dagger - \hat{a}), \quad [\hat{x}, \hat{p}] = 2i.}$$

Consequently $(\Delta x)^2(\Delta p)^2 \geq 1$. For $|\alpha\rangle$:

$$\langle x \rangle = 2\,\mathrm{Re}\,\alpha, \quad \langle p \rangle = 2\,\mathrm{Im}\,\alpha, \quad (\Delta x)^2 = (\Delta p)^2 = 1.$$

For phase angle θ, $\hat{X}_\theta = \hat{a}e^{-i\theta} + \hat{a}^\dagger e^{i\theta}$. Balanced homodyne detection selects θ via the local oscillator. These formulas are convention-sensitive: some texts scale x, p by $\sqrt{2}$.

2. Squeezing changes noise, not the uncertainty bound

For real r, take $\hat{S}(r) = \exp[\frac{r}{2}(\hat{a}^2 - \hat{a}^{\dagger 2})]$. The squeezed vacuum $\hat{S}(r)\,|0\rangle$ has zero means but

$$\boxed{(\Delta x)^2 = e^{-2r}, \qquad (\Delta p)^2 = e^{2r}.}$$

One quadrature is quieter than vacuum and its conjugate noisier; the product stays 1. Squeezing two appropriate modes and combining them at a balanced beam splitter can produce entangled two-mode light.

3. Representing quantum states in phase space

The Husimi function is always positive: $Q(\alpha) = \pi^{-1}\,\langle\alpha|\,\rho\,|\alpha\rangle$. The Glauber–Sudarshan P representation writes $\rho = \int P(\alpha)\,|\alpha\rangle\,\langle\alpha|\,d^2\alpha$; P may be singular or nonclassical. The Wigner function is a real quasiprobability; negative regions provide a nonclassicality witness. These are different representations of the same state. For Gaussian states, a displacement vector $d = \langle R \rangle$, $R = (x, p)^T$, and covariance $V_{jk} = \langle\{\Delta R_j, \Delta R_k\}\rangle/2$ fully specify the state. A Gaussian optical circuit acts as $d' = Sd$, $V' = SVS^T$ (plus environmental noise for a channel).

4. Coherence and photon correlations

$$\boxed{g^{(2)}(0) = \frac{\langle \hat{a}^{\dagger 2}\hat{a}^2 \rangle}{\bar{n}^2} = \frac{\langle n(n-1) \rangle}{\bar{n}^2}.}$$

Coherent light gives 1; single-mode thermal light gives 2; a one-photon Fock state gives 0 (antibunching).

> **Exam method:** identify the quadrature convention; compute first and second moments or $g^{(2)}$; interpret the variance ellipse or photon statistics physically.

Lecture sources: W1L3 pp. 8–27; W3L1 pp. 4–18; W3L2 pp. 3–21; W3L3 pp. 5–28.

Part II. Applications of optics 5 / 15

1. A photon can encode a qubit

A qubit is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $|\alpha|^2 + |\beta|^2 = 1$. A photon's polarisation $|H\rangle, |V\rangle$, two paths $|1,0\rangle, |0,1\rangle$, or two frequency bins can form the logical basis. On the Bloch sphere, $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$ up to an unobservable global phase. Pauli operators are X, Y, Z, and a rotation about axis k is

$$\boxed{R_k(\theta) = e^{-i\theta\sigma_k/2} = \cos(\theta/2)I - i\sin(\theta/2)\sigma_k.}$$

Wave plates implement polarisation rotations; beam splitters and phase shifters implement path-qubit operations. Arbitrary single-qubit unitaries can be decomposed into rotations.

2. Entangled pairs and reduced states

The Bell states are

$$\left|\Phi^{\pm}\right\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \qquad \left|\Psi^{\pm}\right\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}.$$

Although $\left|\Phi^{+}\right\rangle$ is pure,

$$\rho_A = \mathrm{Tr}_B\left(\left|\Phi^{+}\right\rangle\left\langle\Phi^{+}\right|\right) = \tfrac{1}{2}I, \qquad S(\rho_A) = 1 \text{ bit}.$$

Entanglement means the joint state cannot be factored into a product of states on A and B. The reduced state alone does not reveal the correlations.

> **Partial-trace method:** write $\rho_{AB} = \sum_{ijkl}\rho_{ij,kl}|i,j\rangle\langle k,l|$. Then $\mathrm{Tr}_B\,\rho_{AB} = \sum_{i,k,j}\rho_{ij,kj}|i\rangle\langle k|$. For a pure bipartite state, unequal/nonzero Schmidt coefficients signal entanglement.

3. Producing and characterising correlations

Spontaneous parametric down-conversion can generate photon pairs satisfying energy and momentum phase-matching conditions. Appropriate mode selection and interferometry yield polarisation/path entanglement. Entropy $S(\rho) = -\mathrm{Tr}\,\rho\log_2\rho$ quantifies a state's mixedness; for a pure bipartite state, subsystem entropy also measures entanglement. No-cloning forbids a universal unitary that copies an *unknown* arbitrary state.

Lecture sources: W4L1 pp. 3–37; W4L2 pp. 3–28; Week 4 lecture transcripts on photonic qubits and entanglement.

Part II. Applications of optics 6 / 15

1. Teleportation uses entanglement plus classical bits

Alice holds an unknown qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and one half of a shared Bell pair; Bob holds the other half. Alice performs a Bell-basis measurement and sends Bob *two classical bits*. Depending on her outcome, Bob's state is one of $|\psi\rangle, X|\psi\rangle, Z|\psi\rangle, XZ|\psi\rangle$ (up to global phases). He applies the corresponding correction. The original input is destroyed by Alice's measurement; there is no faster-than-light information transfer.

Exam method: write $|\psi\rangle \otimes \left|\Phi^+\right\rangle$; apply CNOT (unknown qubit as control), then Hadamard to the first qubit; regroup by Alice's two measurement bits. Read off Bob's Pauli correction from each branch.

2. Dense coding reverses the resource accounting

If Alice and Bob share a Bell pair, Alice can apply one of I, X, Z, XZ to her half, producing four distinguishable Bell states. She sends *one physical qubit* to Bob, who performs a Bell measurement and recovers *two bits*. The shared entanglement is the essential extra resource.

3. BB84 quantum key distribution

Alice encodes random bits in one of two conjugate bases, such as $\{|H\rangle, |V\rangle\}$ or $\{|+\rangle, |-\rangle\}$, and Bob chooses a random measurement basis. They publicly announce bases, *sift* positions where bases match, estimate error rates using a disclosed sample, then perform error correction and privacy amplification. Security uses measurement disturbance and no-cloning, not secrecy of the basis choices. An intercept-resend eavesdropper choosing random BB84 bases introduces a 25% error rate in the sifted bits in the ideal single-photon model; real security analyses include losses, multi-photon signals, finite-key effects and detector assumptions.

4. Continuous-variable QKD

Rather than discrete photonic qubits, Alice can encode information in optical quadratures; Bob measures them by homodyne/heterodyne detection. An entanglement-based view starts with a two-mode squeezed state and transmission through a noisy/lossy channel. State covariance and channel parameters then describe the observable correlations and security analysis.

Concept check: teleportation transfers an unknown *state*; dense coding transfers two *classical bits*; QKD establishes a *shared secret key*. All have different resource requirements.

Lecture sources: W4L3 pp. 2–35; W4L2 pp. 20–28; Week 4 quantum-communication transcript.

Part II. Applications of optics 7 / 15

1. A noisy optical device is an open quantum system

For a system S interacting with an environment E, evolution is unitary on the *joint* space but generally not on S:

$$\boxed{\rho_S' = \mathcal{E}(\rho_S) = \mathrm{Tr}_E[U(\rho_S \otimes \rho_E)U^\dagger].}$$

The same channel can be expressed with Kraus operators $\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger$, where $\sum_k K_k^\dagger K_k = I$ for trace preservation. The measurement outcome k is ignored here. For pure attenuation, mix the signal a with an environment vacuum mode e on a beam splitter:

$$\hat{a}' = \sqrt{\eta}\,\hat{a} + \sqrt{1-\eta}\,\hat{e}, \quad |1,0\rangle \to \sqrt{\eta}\,|1,0\rangle + \sqrt{1-\eta}\,|0,1\rangle$$

(up to the selected phase convention). Tracing out the environment gives

$$\boxed{\mathcal{E}_\eta(|1\rangle\langle 1|) = \eta\,|1\rangle\langle 1| + (1-\eta)\,|0\rangle\langle 0|\,.}$$

A single-photon superposition loses coherence as well: $\mathcal{E}_\eta(|0\rangle\langle 1|) = \sqrt{\eta}\,|0\rangle\langle 1|$. One must not just replace $\hat{a} \to \sqrt{\eta}\hat{a}$: the vacuum noise operator preserves commutators.

2. Continuous time: Lindblad equation

A Markovian reservoir produces

$$\boxed{\dot{\rho} = -\frac{i}{\hbar}[H,\rho] + \sum_j \gamma_j \mathcal{D}[L_j]\rho}, \quad \mathcal{D}[L]\rho = L\rho L^\dagger - \tfrac{1}{2}\{L^\dagger L, \rho\}.$$

For cavity photon loss take $L = \hat{a}$. The average photon number decays as $\langle n(t)\rangle = \langle n(0)\rangle e^{-\gamma t}$ for zero-temperature damping. A thermal bath requires absorption as well as emission channels.

3. Langevin equation: damping must accompany noise

In the Heisenberg picture a cavity with frequency ω_c and damping rate κ follows a form such as

$$\dot{\hat{a}} = -(i\omega_c + \kappa/2)\hat{a} + \sqrt{\kappa}\,\hat{a}_{\mathrm{in}}(t)$$

with sign convention set by the bath coupling. The input-noise operator is necessary to retain the canonical commutator. In the Gardiner–Collett derivation, a continuum of reservoir modes, a near-resonant/Markov approximation and bath initial conditions lead to both the damping and fluctuation terms.

> **Method:** choose S, E; specify environment state and unitary interaction; apply it to the joint input; trace over E. For time evolution, identify jump operators and rates. Always check $\mathrm{Tr}\,\rho' = 1$.

Lecture sources: W5L1 pp. 7–31; W5L2 pp. 2–25; W2L2 pp. 15–17.

Part II. Applications of optics 8 / 15

1. Postselection changes the quantum operation

If a detector outcome k is retained, the unnormalised state and probability are

$$\tilde{\rho}_k = K_k \rho K_k^\dagger, \qquad p_k = \operatorname{Tr} \tilde{\rho}_k, \qquad \boxed{\rho_k = \tilde{\rho}_k / p_k.}$$

In contrast, $\sum_k \tilde{\rho}_k$ is trace-preserving. For a weakly reflecting beam splitter with a vacuum ancilla, detecting *one* reflected photon implements conditional photon subtraction. For input $|n\rangle$,

$$K_1 |n\rangle = \sqrt{n(1-\eta)}\, \eta^{(n-1)/2} |n-1\rangle$$

up to an irrelevant overall phase. Higher-n amplitudes are preferentially weighted by $\sqrt{n}$; this is *not* the same as deterministically deleting a photon. A no-click conditioning has different photon-number weights.

Exam method: transform the joint input with the beam splitter; project the detector mode onto the specified Fock state; retain the unnormalised signal; calculate p; finally normalise.

2. Jaynes–Cummings: one atom plus one cavity mode

Near resonance, the rotating-wave approximation gives

$$\boxed{H_{\mathrm{JC}} = \hbar\omega_c a^\dagger a + \hbar\omega_a \sigma_+\sigma_- + \hbar g(a\sigma_+ + a^\dagger \sigma_-).}$$

The total excitation number is conserved, so $\{|e,n\rangle, |g,n+1\rangle\}$ form independent two-dimensional subspaces. At resonance their symmetric/antisymmetric dressed-state energies are split by $2\hbar g\sqrt{n+1}$. In the one-excitation manifold the vacuum Rabi doublet has frequency separation $2g$. Away from resonance, diagonalise $\frac{\hbar}{2}\begin{pmatrix} \Delta & 2g\sqrt{n+1} \\ 2g\sqrt{n+1} & -\Delta \end{pmatrix}$ after subtracting a common energy.

3. Tavis–Cummings and bright/dark states

For N identical atoms coupling in phase to a common mode,

$$H_I = \hbar g(aJ_+ + a^\dagger J_-), \quad J_\pm = \sum_i \sigma_\pm^{(i)}.$$

For two atoms, the symmetric one-excitation state $|S\rangle = (|eg\rangle + |ge\rangle)/\sqrt{2}$ couples to $|gg,1\rangle$ with strength $\sqrt{2}g$; the antisymmetric $|A\rangle = (|eg\rangle - |ge\rangle)/\sqrt{2}$ is dark under identical coupling. For a symmetric single excitation in N atoms, the coupling is $g\sqrt{N}$. This collective enhancement underlies superradiant behaviour, though collective coherent exchange and spontaneous superradiant decay are not identical phenomena.

Lecture sources: W5L3 pp. 3–25; W6L1 pp. 5–29; W6L2 pp. 5–25.

Part II. Applications of optics 9 / 15

1. Parameter estimation and Fisher information

A probe accumulates an unknown phase ϕ through $|\psi_\phi\rangle = e^{-i\phi G}|\psi_0\rangle$. For a chosen measurement with outcomes x, classical Fisher information is

$$F(\phi) = \sum_x p(x|\phi)\left[\partial_\phi \ln p(x|\phi)\right]^2 .$$

For M independent repetitions, an unbiased estimator obeys the Cramér–Rao bound $\mathrm{Var}(\tilde{\phi}) \geq 1/[MF(\phi)]$. Optimising over physically allowed measurements gives the quantum Fisher information F_Q:

$$\boxed{\mathrm{Var}(\tilde{\phi}) \geq \frac{1}{MF_Q}, \qquad F_Q = 4(\Delta G)^2 \quad \text{for a pure probe.}}$$

If phase is generated by photon number $G = \hat{n}$, coherent light has $F_Q = 4\bar{n}$. The standard deviation therefore scales as $1/\sqrt{\bar{n}}$ (shot-noise scaling, up to constant factors). Carefully chosen nonclassical states can offer better scaling under specified resource and loss assumptions.

2. Interferometers convert phase to intensity

A Mach–Zehnder interferometer uses two beam splitters separated by a relative phase. In a balanced convention, a photon entering one input has output probabilities

$$P_1 = \cos^2(\Delta\phi/2), \qquad P_2 = \sin^2(\Delta\phi/2)$$

up to which port is named 1 or 2. The phase shift is invisible to direct intensity measurement on a single path until the two arms recombine. A dark output at $\Delta\phi = 0$ becomes bright when the relative phase changes.

3. Why squeezing helps

A coherent state has equal vacuum noise in every quadrature. If the measured quadrature is squeezed below vacuum noise, the same signal displacement can be resolved with smaller fluctuations. This underlies squeezed-light enhancement of precision interferometers, including gravitational-wave detectors. Real gain depends on optical loss, mode matching and measurement angle.

> **Typical derivation:** identify the encoded generator G; calculate $\langle G\rangle, \langle G^2\rangle$; find F_Q, then the bound for M repetitions. For a concrete detector, compute $p(x|\phi)$ first and compare its classical F to F_Q.

Lecture sources: W6L3 pp. 5–37; W3L2 on squeezing; W2L2 on beam splitters.

Part III. Atomic physics 10 / 15

1. Start from a Coulomb-bound electron

For a hydrogen-like ion of nuclear charge Z, the nonrelativistic energy, with reduced mass μ, is

$$\boxed{E_n = -\frac{\mu Z^2 e^4}{2(4\pi\epsilon_0)^2\hbar^2 n^2} \simeq -13.6\,\text{eV}\,\frac{\mu}{m_e}\frac{Z^2}{n^2}.}$$

The eigenstates separate into radial and angular factors:

$$\psi_{nlm}(\mathbf{r}) = R_{nl}(r)Y_{lm}(\theta,\phi), \qquad \int_0^\infty |R_{nl}(r)|^2 r^2\,dr = 1.$$

With $u_{nl} = rR_{nl}$, the radial Schrödinger equation takes a one-dimensional form,

$$-\frac{\hbar^2}{2\mu}\frac{d^2u}{dr^2} + \left[V(r) + \frac{\hbar^2 l(l+1)}{2\mu r^2}\right]u = Eu.$$

The angular-momentum barrier suppresses high-l wavefunctions near the nucleus. Hydrogenic orbitals have $l = 0,\ldots,n-1$, $m_l = -l,\ldots,l$, and orbital parity $(-1)^l$. A transition photon has frequency $\nu = (E_i - E_f)/h$ and wavelength $hc/(E_i - E_f)$.

2. Many-electron atoms: the central-field approximation

For more than one electron, electron–electron repulsion breaks the simple n-only degeneracy:

$$H = \sum_i \left[\frac{p_i^2}{2m_e} - \frac{Ze^2}{4\pi\epsilon_0 r_i}\right] + \sum_{i<j}\frac{e^2}{4\pi\epsilon_0 r_{ij}}.$$

An effective central potential replaces part of the many-body interaction by screening. Hartree–Fock improves this by self-consistently including direct and exchange effects in a Slater determinant. The Aufbau $n + l$ ordering is a useful rule of thumb, not an exact theorem; shell energies and order change with ionisation and configuration.

3. Nuclear size and relativistic effects

A finite nuclear charge distribution changes the potential at small r, especially affecting orbitals that penetrate the nucleus. In first-order perturbation theory,

$$\boxed{\Delta E^{(1)} = \langle\psi|\,\Delta V\,|\psi\rangle = \int_0^\infty |R_{nl}(r)|^2 \Delta V(r)\,r^2\,dr.}$$

The sign and magnitude depend on the defined $\Delta V = V_{\text{finite}} - V_{\text{point}}$. For heavy atoms, spin–orbit and other relativistic corrections grow important; the Dirac equation incorporates the electron's spin and relativistic kinematics from the outset.

Method: specify the central potential and boundary conditions; choose R or $u = rR$ consistently; solve energies/states; evaluate corrections using normalised wavefunctions and check units.

Lecture sources: Week 8 slides pp. 2–7, 26–36; Week 8 atomic transcripts; nuclear-size assignment.

Part III. Atomic physics 11 / 15

1. Adding angular momenta

Operators $\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2$ obey

$$\hat{J}^2 \left|jm\right\rangle = \hbar^2 j(j+1) \left|jm\right\rangle, \quad \hat{J}_z \left|jm\right\rangle = \hbar m \left|jm\right\rangle,$$

$$\boxed{j = |j_1 - j_2|,\ |j_1 - j_2| + 1, \ldots, j_1 + j_2.}$$

In atomic LS coupling, first sum the electron orbital momenta to L and electron spins to S, then form $J = L + S$. The term notation is ${}^{2S+1}L_J$, with $L = 0, 1, 2, 3, 4 \leftrightarrow S, P, D, F, G$. For hyperfine structure $\mathbf{F} = \mathbf{J} + \mathbf{I}$ where I is nuclear spin. A multiplet with angular momentum j contains $2j + 1$ projections.

2. Equivalent electrons and Hund's rules

For nd^2, there are 10 single-electron spin-orbitals and $\binom{10}{2} = 45$ antisymmetric two-electron states. Allowed LS terms are

$$^3F_{2,3,4}, \quad ^3P_{0,1,2}, \quad ^1G_4, \quad ^1D_2, \quad ^1S_0.$$

The check $21 + 9 + 9 + 5 + 1 = 45$ tests completeness. Hund's rules (in their range of validity): maximise S, then maximise L; in a less-than-half-filled subshell, the lowest J of a given term lies lowest. This predicts a ${}^3F_2$ ground state for a simple d^2 system. It does not determine an unambiguous ordering among all excited terms.

3. Wigner–Eckart factors out the geometry

A rank-k spherical tensor has components $q = -k, \ldots, k$. For a fixed reduced-matrix-element convention,

$$\boxed{\langle j' m' | T^{(k)}_q | jm \rangle = (-1)^{j'-m'} \begin{pmatrix} j' & k & j \\ -m' & q & m \end{pmatrix} \langle j' \| T^{(k)} \| j \rangle.}$$

For a nonzero $3j$ symbol, magnetic quantum numbers must sum to zero, each $|m_i| \le j_i$, and the triangle rule must hold. In this arrangement $m' = m + q$. The $3j$ coefficient describes geometry; the reduced element encodes the actual atom/operator.

Short example: adding $j_1 = 0$ to $j_2 = 1$ has CG coefficient $\langle 0, 0; 1, 0 | 1, 0 \rangle = 1$. Using the CG–$3j$ conversion yields $\begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} = -1/\sqrt{3}$. Check zeros *before* doing an explicit sum.

Lecture sources: Week 8 slides pp. 7–19; Week 8 transcripts on LS coupling, 3-j symbols and Wigner–Eckart.

Part III. Atomic physics 12 / 15

1. Fine structure and hyperfine structure

In a spin–orbit model, $H_{\mathrm{SO}} \propto \mathbf{L}\cdot\mathbf{S}$. Since $\mathbf{J}=\mathbf{L}+\mathbf{S}$,

$$\langle \mathbf{L}\cdot\mathbf{S}\rangle = \frac{\hbar^2}{2}[J(J+1)-L(L+1)-S(S+1)].$$

This separates J levels within a fixed LS term. For the magnetic-dipole hyperfine interaction,

$$H_{\mathrm{hfs}} = \frac{A}{\hbar^2}\mathbf{I}\cdot\mathbf{J}, \qquad \boxed{E_F = \frac{A}{2}[F(F+1)-I(I+1)-J(J+1)].}$$

Here $F=|I-J|,\ldots,I+J$; each level has $2F+1$ projections when no magnetic field breaks degeneracy. Electric-quadrupole hyperfine terms can also appear for sufficient nuclear/electronic angular momentum.

Worked example: $I=5/2, J=1/2$ gives $F=2,3$. Using $I(I+1)=35/4,\ J(J+1)=3/4$,

$$\boxed{E_2=-\frac{7A}{4},\quad E_3=\frac{5A}{4},\quad \Delta E = 3A.}$$

The multiplicities $5+7=12=(2I+1)(2J+1)$. For $A<0$, the energy order is reversed.

2. Magnetic and electric fields

In a weak magnetic field, F, m_F are approximately good quantum numbers. The linear Zeeman shift is

$$\Delta E \simeq g_F \mu_B m_F B, \quad g_F \simeq g_J \frac{F(F+1)+J(J+1)-I(I+1)}{2F(F+1)}$$

for $F>0$ and neglecting the small nuclear magnetic term. At strong fields, I and J decouple approximately, so a basis $|Jm_J, Im_I\rangle$ becomes useful. A static electric field induces a Stark shift. For a nondegenerate parity eigenstate with no permanent dipole,

$$\Delta E^{(2)}_{\mathrm{Stark}} = -\tfrac{1}{2}\alpha(0)\mathcal{E}^2, \qquad \alpha(0) = 2\sum_{n\neq i}\frac{|\langle n|d_z|i\rangle|^2}{E_n-E_i}$$

with field along z (and compatible dipole conventions). Degenerate states require diagonalisation rather than this simple nondegenerate formula.

3. Isotope shifts

Changing neutron number changes the nuclear mass and charge distribution. Spectral isotope shifts reflect normal/specific mass shifts and field (volume) shifts; these can reveal nuclear radius differences when atomic sensitivity coefficients are known.

Method: choose the coupling/field regime, list allowed F, compute splitting; for a perturbation, identify the operator and apply first- or second-order theory.

Lecture sources: Week 8 slides on relativistic structure; Week 9 pp. 2–7, 21–27; hyperfine tutorial and transcripts.

Part IV. Applications of atomic physics 13 / 15

1. Optical transitions: identify the allowed operator

An electromagnetic transition has both angular and parity constraints. Electric multipole Ek: rank k, parity $(-1)^k$. Magnetic multipole Mk: rank k, parity $(-1)^{k+1}$. For a transition $J, m \to J', m'$, angular momentum requires

$$\boxed{|J - J'| \le k \le J + J', \qquad m' = m + q,\ |q| \le k.}$$

Type	Parity	Usual ΔJ	Single-electron orbital check
E1	changes	$0, \pm 1$, not $0 \leftrightarrow 0$	$\Delta l = \pm 1$
M1	same	$0, \pm 1$, not $0 \leftrightarrow 0$	typically $\Delta l = 0$
E2	same	$0, \pm 1, \pm 2$, triangle rule	$\Delta l = 0, \pm 2$, triangle rule

Total L, S rules and level mixing depend on the coupling approximation. *Example:* $s \to p$ can be E1; $s \to d$ can be E2; $s \to p$ cannot be E2 because the orbital parity changes.

2. Einstein coefficients, linewidths, lifetimes

Stimulated absorption/emission depend on the incident spectral energy density; spontaneous decay occurs even in vacuum. For an E1 transition between specified states (with consistent treatment of polarisation/sublevels),

$$\boxed{A_{if} = \frac{\omega_{if}^3}{3\pi\epsilon_0\hbar c^3}|\langle f|\mathbf{d}|i\rangle|^2, \quad \tau_i = \Big(\sum_{f:E_f<E_i} A_{if}\Big)^{-1}.}$$

Here $\mathbf{d} = -e\mathbf{r}$. For $R(r)$ normalisation, the radial dipole integral is $\int_0^\infty R_f^*(r) R_i(r) r^3\, dr$; multiply by the angular factor or reduced matrix element. The natural linewidth (angular frequency) is of order $\Gamma = \sum_f A_{if} = 1/\tau$. With $\hbar\omega$ in atomic units, convert energies, dipoles and time consistently ($t_{\text{au}} = 2.418884 \times 10^{-17}$ s).

3. Laser cooling and trapping

Absorbing a photon transfers momentum $+\hbar\mathbf{k}$; spontaneous emission gives a random recoil. For a driven two-level atom, the scattering rate is

$$R_{\text{sc}} = \frac{\Gamma}{2}\frac{s}{1 + s + (2\Delta/\Gamma)^2}, \quad \mathbf{F}_{\text{sc}} = \hbar\mathbf{k}R_{\text{sc}}, \quad s = I/I_{\text{sat}}.$$

The atom experiences Doppler-shifted detuning $\Delta_{\text{eff}} = \Delta - \mathbf{k}\cdot\mathbf{v}$. Opposing red-detuned beams yield velocity-dependent damping (optical molasses). A basic two-level Doppler limit is $T_D = \hbar\Gamma/(2k_B)$. In a magneto-optical trap a magnetic-field gradient plus polarisation-dependent Zeeman shifts adds a restoring force.

> **Method:** check selection rules first; obtain matrix elements and rates, then lifetime. For cooling problems, identify detuning, Doppler shift, scattering momentum and restoring/damping force.

Lecture sources: Week 9 pp. 8–21; Week 10 pp. 4–58 (radiation), 60–130 (optical pumping/cooling).

Part IV. Applications of atomic physics 14 / 15

1. Why an atom can test fundamental physics

Electronic energies and transition amplitudes are calculable using atomic structure theory and can be measured with very high precision. Comparing them tests electromagnetic theory, the weak interaction and possible physics beyond the Standard Model. Week 12 develops two complementary examples: atomic parity violation (APV) and electric dipole moments (EDMs).

2. Atomic parity violation: weak interactions mix parity

Under parity P, $\mathbf{r} \to -\mathbf{r}$. The electric dipole operator $\mathbf{d} = -e\mathbf{r}$ is parity-odd, so an E1 matrix element between same-parity states vanishes when parity is conserved. The weak neutral-current interaction contains parity-violating electron–nucleus couplings, so

$$H = H_{\mathrm{EM}} + H_{\mathrm{weak}}, \qquad \left|\tilde{i}\right\rangle \simeq |i\rangle + \sum_{n\neq i} |n\rangle \, \frac{\langle n|H_{\mathrm{weak}}|i\rangle}{E_i - E_n}.$$

To first order, two nominally same-parity states may acquire a small parity-violating E1 amplitude:

$$\boxed{E_{\mathrm{PV}} \simeq \sum_n \left[\frac{\langle f|d|n\rangle\langle n|H_W|i\rangle}{E_i - E_n} + \frac{\langle f|H_W|n\rangle\langle n|d|i\rangle}{E_f - E_n} \right].}$$

The weak charge is approximately $Q_W \simeq -N + Z(1 - 4\sin^2\theta_W)$ at tree level, with electroweak radiative corrections and nuclear-structure details needed for precision predictions. APV experiments combine precision spectroscopy, electronic-structure theory and weak-interaction physics.

3. EDMs probe P- and T-violating physics

An intrinsic permanent electric dipole aligned with angular momentum reverses under parity and (for spin alignment) under time reversal. In an applied electric field,

$$\boxed{H_{\mathrm{EDM}} = -\mathbf{d}\cdot\boldsymbol{\mathcal{E}}.}$$

Comparing spin/precession energy under reversal of electric and magnetic fields seeks an EDM-like shift. A nonzero fundamental EDM at an accessible scale could indicate CP-violating physics beyond the Standard Model, subject to the relationship between T and CP symmetries. Schiff screening suppresses the electric field at point-like constituents of an ideal nonrelativistic neutral atom in electrostatic equilibrium. Relativistic effects, finite nuclear size and molecular enhancement allow sensitive electron/nuclear EDM measurements.

Exam method: determine whether the operator is even/odd under P or T; identify the small perturbation; use first-order state mixing and a matrix element. Explain what observable is measured and why a symmetry-forbidden background is suppressed.

Lecture sources: Week 12 pp. 2–55 (weak interaction/APV), 65–129 (EDM, screening and experimental systems).

Part IV. Applications of atomic physics 15 / 15

1. Atomic clocks measure a stable transition frequency

A clock locks a local oscillator to a narrow atomic transition. The modern SI second is defined by the unperturbed ground-state hyperfine transition frequency of ^{133}Cs:

$$\boxed{\nu_{\mathrm{Cs}} = 9\,192\,631\,770\ \mathrm{Hz}.}$$

Cs has configuration [Xe]$6s$, electronic $J = 1/2$, nuclear $I = 7/2$, hence $F = 3, 4$. The clock transition is

$$\boxed{|F = 3, m_F = 0\rangle \longleftrightarrow |F = 4, m_F = 0\rangle\,,}$$

not $F = 3 \to F = 3$. The $m_F = 0$ choice strongly reduces first-order Zeeman sensitivity near zero field. Optical clocks exploit much higher carrier frequencies, e.g. strontium clock transitions around 429 THz, and narrow, often nominally forbidden, transitions. Trap/lattice configurations help reduce motion and frequency shifts.

2. Linewidth, interrogation time and stability

The quality factor is $Q = \nu_0/\Delta\nu$. Long-lived excited states have narrow natural linewidths; long coherent interrogation time T can produce spectral features with width of order $1/T$. For N independent atoms, projection noise typically scales as $1/\sqrt{N}$ and repeated measurements improve statistics as the square root of averaging time under appropriate noise models. Real clock accuracy requires controlling black-body shifts, Zeeman and Stark shifts, collisions, gravitational redshift and laser noise. Clock comparisons can test relativity and variation of fundamental constants.

3. Find the right formula quickly

Question type	Starting formula or test	Page
Photon counts	$p_n,\ \bar{n},\ (\Delta n)^2,\ g^{(2)}$	3–4
Coherent/squeezed light	$\hat{a}\,\lvert\alpha\rangle = \alpha\,\lvert\alpha\rangle\,,\ (\Delta x)^2 = e^{-2r}$	2,4
Beam splitter / HOM	transform $\hat{a}^\dagger, \hat{b}^\dagger$	3
Photonic entanglement	Bell basis; partial trace / entropy	5
Teleport / QKD	measurement branches; basis sifting	6
Optical loss	joint unitary then Tr_E	7
Photon subtraction	project detector mode, then normalise	8
Cavity coupling	JC excitation block; dressed eigenvalues	8
Interferometric sensing	$F_Q = 4\,\mathrm{Var}(G)$	9
Radial / finite nucleus	$u = rR,\ \Delta E = \langle\Delta V\rangle$	10
Term symbols and $3j$	triangle rule, microstate count, W–E	11
Hyperfine / Zeeman / Stark	$E_F,\ g_F m_F B,\ -\alpha E^2/2$	12
E1 / M1 / E2 and lifetimes	rank + parity + A_{if}, then τ	13
Atomic PV / EDM	weak-induced mixing / $-\mathbf{d}\cdot\mathbf{E}$	14
Clocks	F, m_F choice; $\nu_0/\Delta\nu$	15

Universal exam workflow: recognise the model and basis; write assumptions/conventions; derive or quote the governing operator; calculate with explicit normalisation; check dimensions, limiting cases, parity/selection rules and degeneracy; interpret the observable physically.

Lecture sources: Week 13 pp. 1–35; Week 9 on hyperfine shifts; Week 10 on linewidths and forbidden lines.